Battery Chemistries for Grid-Scale Energy Storage: A Comparative Study

Abstract

This paper compares three lithium and sodium battery chemistries for stationary storage: lithium iron phosphate (LFP), nickel manganese cobalt (NMC) and sodium-ion. Cells were cycled for 3,000 cycles. LFP retained 92% of its capacity, sodium-ion 86% and NMC 81%.

1. Introduction

Grid operators need storage that lasts 15 to 20 years at a low cost per stored kilowatt-hour. Energy density matters less than in electric vehicles, while cycle life, cost and safety matter more.

2. Methodology

Twelve cells of each chemistry were cycled at 1C charge and discharge between 10% and 90% state of charge in a climate chamber held at 25 °C.

Capacity was measured every 500 cycles with a slow C/3 reference test. End of life was defined as the point where a cell retains 80% of its initial capacity.

3. Capacity retention

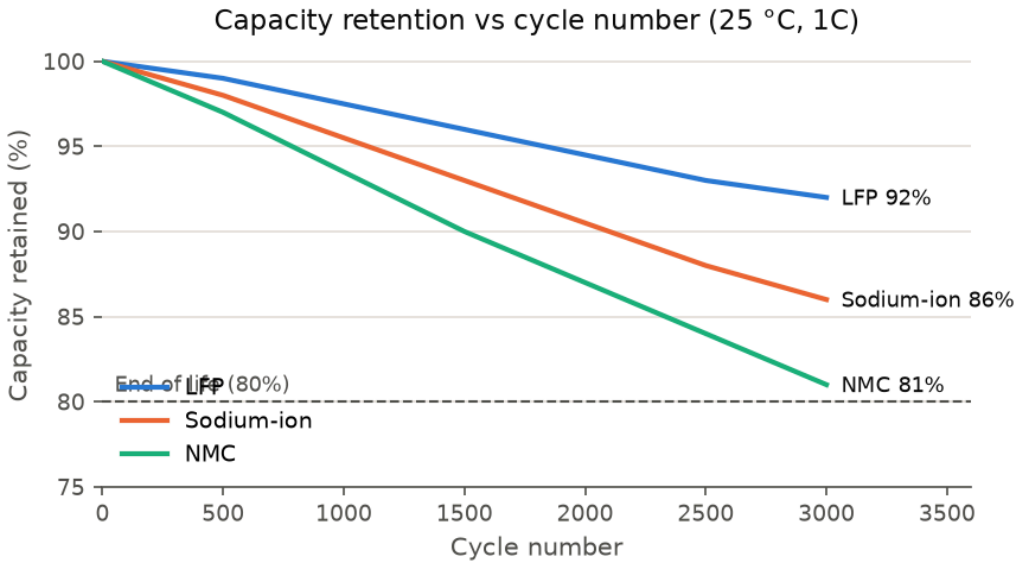


Figure 3: Capacity retention versus cycle number for LFP, NMC and sodium-ion cells.

NMC degraded fastest and is projected to reach the end-of-life threshold at about 3,200 cycles. LFP showed the slowest and most linear capacity fade.

4. Cost and performance comparison

Property	LFP	NMC	Sodium-ion
Energy density (Wh/kg)	165	240	140
Cycle life to 80% (cycles)	6,000	3,200	4,500
Cell cost (USD/kWh)	95	120	85
Thermal runaway onset (C)	270	210	300

NMC has the highest energy density but the shortest cycle life and the lowest thermal runaway temperature. Sodium-ion is the cheapest chemistry and the most resistant to thermal runaway.

5. Conclusions

LFP offers the best balance of cycle life, safety and cost for stationary storage today. We recommend LFP for new 4-hour storage systems and suggest re-evaluating sodium-ion in 2027, when larger production volumes are expected to improve its energy density.